

Short-Term Traffic Flow Forecasting Using Sampling Markov Chain Method with Incomplete Data

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Abstract—Short-term traffic flow forecasting is an important problem in the research area of intelligent transportation system. In practical situations, flow data may be incomplete, that is, partially missing or unavailable, where few methods could implement forecasting successfully. A method called Sampling Markov Chain is proposed to deal with this circumstance. In this paper, the traffic flow is modeled as a high order Markov Chain; and the transition probability from one state to the other state is approximated by Gaussian Mixture Model (GMM) whose parameters are estimated with Competitive Expectation Maximum (CEM) algorithm. The incomplete data in forecasting the trend of Markov Chain is represented by enough points sampled using the idea of Monte Carlo integration. Experimental results show that the Sampling Markov Chain method is applicable and effective for short-term traffic flow forecasting in case of incomplete data.

I. INTRODUCTION

In recent years, Intelligent Transportation Systems (ITS) have achieved great developments. However, many problems, including traffic management and congestion control, still remain unsolved. Precise analysis of historical trends of traffic flows or transportation forecasting therefore becomes fundamental tasks in order to progress successfully and smoothly in our daily routines.

Short-term traffic flow forecasting, which is to determine the traffic volume in the next time interval usually in the range of five minutes to half an hour, is an important problem in the research area of ITS. In the past years, some approaches ranging from simple to complex were proposed on this theme. Simple ones, such as random walk (which is informed only by the current conditions), historical average (whose prediction is solely based on the average of all corresponding observed flow rates), and informed historical average (which combines the above two methods) can only work well in specific situations [1]. One class of the complex approaches, UTCS (Urban Traffic Control System) prediction method is based on forecasting philosophies similar to the above methods, and it would either bring along similar drawbacks or become highly complex [1]. There are also other elaborate methods including approaches based on time series models (including ARIMA, seasonal ARIMA), Kalman filter theory, neural network approaches, non-parametric methods, simulation models,

local regression models, layered models and Markov Chain models [1]~[7].

Although these methods have alleviated difficulties in data modeling and forecasting to a large extent, they can't work well while the data used for forecasting is incomplete, that is, partially missing or unavailable. There are usually two methods to deal with this case. One is filling up the incomplete data with their historical average and the other is screening out the incomplete data completely. The forecasting performance of the first approach is very limited and the second method often leads to the impossibility of forecasting.

In this paper, we propose a new method called Sampling Markov Chain by modeling the traffic flow as a Markov Chain and integrating the idea of Monte Carlo integration to carry out forecasting with incomplete data. Experimental results show that our method is applicable and effective for short-term traffic flow forecasting in case of incomplete data.

The remainder of the article is organized as follows. In section 2, we show the Markov Chain model of traffic flow and the application of Monte Carlo integration, and then achieve the formulation of Sampling Markov Chain method for optimal forecasting under the criterion of Minimum Mean Square Error (M.M.S.E.). Besides, the Gaussian Mixture Model (GMM) used to approximate the transition probability of Markov Chain states and the Competitive Expectation Maximum (CEM) algorithm used to estimate the parameters of GMM are also explained. Experiments in section 3 are implemented to verify the applicability and effectiveness of our Sampling Markov Chain method with real traffic data obtained from UTC/SCOOT System in Beijing city. Then, in section 4, we sum up our method and give some useful discussions about the orientation of future research.

II. METHOD

One important characteristic of traffic flow is that the traffic volume to be forecasted in the next time interval has strong but not deterministic relationship to the current and recent states. Markov Chain model has been successfully applied to describe this characteristic of traffic flow, where the state to be forecasted is considered to obey a probability distribution and the probability of the next state is determined by the current and immediately preceding states [7]. Here we adopt this idea and use Markov Chain model to describe the traffic flow.

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A. Markov Chain model for traffic flow

An N-order Markov Chain is employed to describe a series which has the following attribution: given the current and N-1 preceding states, the future state is independent to the states prior to the given states. If we denote $x(t)$ as the state of the traffic flow at time interval t , then the probability density function $p(Y|X) = p(x(t+1)|x(t), x(t-1), \dots, x(t-N+1))$ describes the transition probability of $x(t+1)$ given the current and N-1 previous states of traffic flow, where multidimensional random vector $X = [x(t), x(t-1), \dots, x(t-N+1)]^T$, random variable $Y = x(t+1)$ and N is the order of Markov Chain Model. If $p(Y|X)$ is known, then the optimal estimation of Y would be given as

$$\hat{Y} = E(Y|X) \quad (1)$$

under the criterion of Minimum Mean Square Error (M.M.S.E.) [7].

If we obtain $p(Y, X)$, then $p(Y|X)$ can be easily got by the Bayesian Theorem. That is

$$p(Y|X) = \frac{p(Y, X)}{p(X)} = \frac{p(Y, X)}{\int p(Y, X) dY}$$

B. GMM and the parameter estimation

How to deduce $p(Y, X)$ with the given data is a problem. In this paper the joint probability distribution $p(Y, X)$ is approximated through a Gaussian Mixture Model (GMM) which can approximate an arbitrary probability density function with enough accuracy. We prefer the Gaussian Mixture model mainly based on the following three reasons:

- 1). Many events or phenomena in the natural world *per se* obey Gaussian distributions.
- 2). The Gaussian function has its convenience in mathematical processing which can be seen below.
- 3). We can approximate an arbitrary probability distribution with the combination of sufficient Gaussian functions.

The GMM representation of $p(Y, X)$ is as follows:

$$p(Y; X) = \sum_{m=1}^M \alpha_m G(Z; \mu_m, \Sigma_m) \quad (2)$$

where $Z = [Y \ X]$ is multiple random variable. $G(Z; \mu_m, \Sigma_m)$ denotes the m th component of the GMM, which is a Gaussian density function with mean μ_m and covariance matrix Σ_m . α_m is the mixing coefficient of the m th component. M is the number of components.

Because of its appealing accuracy and low requirement for the amount of training data, usually we use Maximum Likelihood Estimation (MLE) to carry out parameter estimation of α_m , μ_m , and Σ_m with given data. Although the EM algorithm is an effective method to carry out MLE, it usually converges to local maxima. To find a global maximum is more significant in most cases. The CEM algorithm overcomes the drawbacks of basic EM algorithm; besides, it is also capable of automatically choosing the

number of mixing components and is insensitive to initial configuration of the number of the mixture components and model parameters [8]. Therefore, we use CEM algorithm to carry out MLE in this paper.

If we rewrite $\mu_m = [\mu_{my}, \mu_{mx}]^T$ and $\Sigma_m = \begin{bmatrix} \Sigma_{myy} & \Sigma_{myx} \\ \Sigma_{myx} & \Sigma_{mxx} \end{bmatrix}$, we can get

$$\begin{aligned} p(Y, X) &= \sum_{m=1}^M \alpha_m G(Z; \mu_m, \Sigma_m) \\ &= \sum_{m=1}^M \alpha_m G(X; \mu_{mx}, \Sigma_{mxx}) G(Y; \mu_{my|x}, \Sigma_{my|x}), \end{aligned}$$

and the conditional probability density function $p(Y|X)$ would be

$$p(Y|X) = \sum_{m=1}^M \beta_m G(Y; \mu_{my|x}, \Sigma_{my|x}) \quad (3)$$

where

$$\begin{aligned} \beta_m &= \frac{\alpha_m G(X; \mu_{mx}, \Sigma_{mxx})}{\sum_{j=1}^M \alpha_j G(X; \mu_{jx}, \Sigma_{jxx})} \\ \mu_{my|x} &= \mu_{my} - \Sigma_{myx} \Sigma_{mxx}^{-1} (\mu_{mx} - X) \\ \Sigma_{my|x} &= \Sigma_{myy} - \Sigma_{myx} \Sigma_{mxx}^{-1} \Sigma_{mxy} \end{aligned}$$

With this model, we can neatly get the optimal estimation of Y given X . The optimal forecasting $\hat{Y}$ can be represented as the following form:

$$\begin{aligned} \hat{Y} &= E(Y|X) \\ &= \int Y p(Y|X) dY \\ &= \sum_{m=1}^M \beta_m \int Y G(Y; \mu_{my|x}, \Sigma_{my|x}) dY \\ &= \sum_{m=1}^M \beta_m \mu_{my|x}. \end{aligned} \quad (4)$$

Now if $x(t)$ in X is missing, then how can we forecast Y ? This is what we will mainly talk about in the following section.

C. Sampling Markov Chain method

Let X_{N-1} denote $x(t-1), x(t-2), \dots, x(t-N+1)$, then we have $E(Y|X) = E(Y|x(t), X_{N-1})$ and the optimal forecasting of Y in (4) can't be easily calculated because of missing data $x(t)$. Because $x(t-1), x(t-2), \dots, x(t-N)$ can determine $x(t)$ statistically in our N-order Markov Chain model assumption, we can take it for granted that $x(t-1), x(t-2), \dots, x(t-N)$ and $x(t-2), \dots, x(t-N+1)$ together determine $x(t+1)$, that is $x(t-1), x(t-2), \dots, x(t-N)$ determine $x(t+1)$ in the statistical sense. We can deduce the formulation of $p(Y|x(t-1), \dots, x(t-N))$ as follows (here we assume the data for $x(t-1), x(t-$

2), ..., $x(t - N)$ is intact when $x(t)$ is missing).

$$\begin{aligned}
& p(Y|x(t-1), \dots, x(t-N)) \\
&= \int p(Y, x(t)|x(t-1), \dots, x(t-N))dx(t) \\
&= \int p(Y|x(t), \dots, x(t-N)) \cdot \\
&\quad p(x(t)|x(t-1), \dots, x(t-N))dx(t) \\
&= \int p(Y|X)p(x(t)|x(t-1), \dots, x(t-N))dx(t) \quad (5)
\end{aligned}$$

Monte Carlo integration tells us, for a random variable or random vector Z with distribution $\pi(\cdot)$, $E[f(Z)]$ can be evaluated as $E[f(Z)] = \int f(z)\pi(z)dz \approx \frac{1}{n} \sum_{i=1}^n f(Z_i)$, where $\{Z_i, i = 1, \dots, n\}$ are samples drawn from $\pi(\cdot)$ [9].

Considering (5), we have $p(Y|x(t-1), \dots, x(t-N)) \approx \frac{1}{n} \sum_{i=1}^n p(Y|X_{ti})$, where $X_{ti} = \{x(t_i), x(t-1), \dots, x(t-N+1)\}$. Then the optimal forecasting of Y with missing data $x(t)$ can be given as

$$E(Y|x(t-1), \dots, x(t-N)) = \frac{1}{n} \sum_{i=1}^n E(Y|X_{ti}), \quad (6)$$

where $E(Y|X_{ti})$ can be obtained from (4).

III. EXPERIMENTS

The experimental data for analysis is the vehicular flow rates of discrete time series recorded every 15 minutes in March, 2002 by the UTC/SCOOT system in Traffic Management Bureau of Beijing city, whose unit is vehicles per hour (veh/h). Traffic flows in 26 days are used in our experiments. Among these days, the former 22 days data are used to learn the transition probability density function $p(Y|X)$ and the residual data of 4 days are employed to test the forecasting performance of the Sampling Markov Chain method. We have done experiments for four different road sections which are randomly selected, and the experimental results are listed in Table 1.

The flow chart of our forecasting procedure can be described as follows.

- 1). Approximate the joint probability distribution $p(Y, X)$ by GMM and CEM algorithm using the methods explained in section 2.1 and section 2.2.
- 2). Deduce the conditional probability distribution of the incomplete data and the given intact data, and then sample the incomplete points from this distribution.
- 3). Carry out the optimal estimation of Y in the form of (6).

Here, we take road section Liangmaqiao as an example to show our approach. The raw data of road section Liangmaqiao in 26 days is given in Fig. 1.

In modeling the traffic flow, we use a 4-order Markov Chain model ($N=4$). Thus, based on the training data we can approximate the joint probability $p(Y, X)$ by Gaussian mixture model whose parameters are estimated with CEM algorithm. In Fig. 2 the first two dimensions of state vector

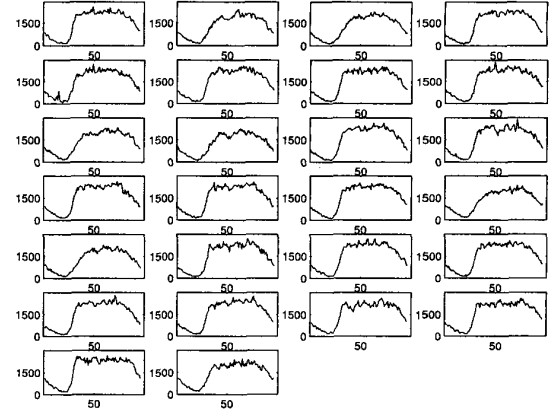


Fig. 1. Traffic flow of Liangmaqiao in 26 days

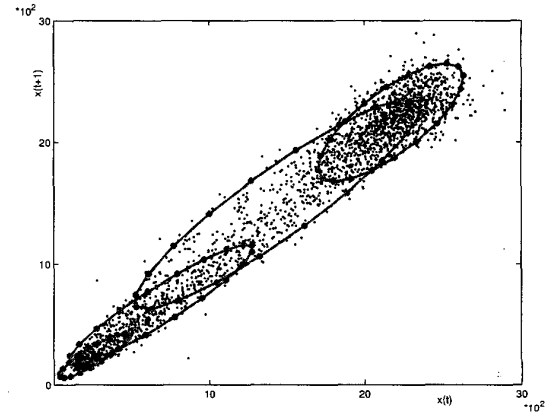


Fig. 2. The training data and the estimated GMM

$[Y, X]$ are shown with one point corresponding to one sample in the training data set. The ellipse is corresponding to the estimated Gaussian component with the center and shape denoting the estimated parameters. From the figure, we can see that the component number of GMM is selected as 4 automatically by the CEM algorithm.

As a matter of convenience, the transition probability in the Markov Chain is assumed to be time independent, so $p(x(t)|x(t-1), \dots, x(t-N)) = p(x(t+1)|x(t), \dots, x(t-N+1))$. When data $x(t)$ is missing, we can use the idea of Monte Carlo integration to sample points from the distribution $p(x(t)|x(t-1), \dots, x(t-N))$. As this probability distribution is a mixture of Gaussian distribution deduced from $p(Y, X)$, we can carry out sampling easily. Given the traffic volumes of $x(t-1), \dots, x(t-N)$, we draw 400 points (which are enough for our problem) from the distribution. Then we use (6) to forecast $x(t+1)$.

The four days' forecasted flow vs. observed flow is shown in Fig. 3 where the dotted star line shows the forecasted data while the solid line shows the observed data.

The above procedure deals with the case when $x(t)$ is

missing in forecasting $x(t+1)$. We also carry out the cases when $x(t-1)$, $x(t-2)$, ..., and $x(t-N+1)$ are missing respectively. The forecasting error in Table 1 is the average of these four corresponding errors.

In order to evaluate our Sampling Markov Chain method, we also design two other methods called Historical Average Markov Chain method and Joint Distribution Recalculating method to deal with the incomplete traffic flow forecasting with the same data. The Historical Average Markov Chain method treats the missing data as the average of its historical value according to its time in a day, and then using the Markov Chain method to forecast. The Joint Distribution Recalculating method is to compute joint distribution $p(x(t+1), x(t-1), \dots, x(t-N))$ instead of $p(x(t+1), x(t), x(t-1), \dots, x(t-N+1))$, and then use the expectation $E(x(t+1)|x(t-1), \dots, x(t-N))$ to forecast $x(t+1)$ directly. This method might need to compute a distribution of much higher dimension, e.g. when $x(t-3)$ is missing, it should compute the distribution $p(x(t+1), \dots, x(t-2), x(t-4), \dots, x(t-7))$.

Besides, we also carry out forecasting with methods of Historical Average, Random Walk and Markov Chain model (used in [7]) with the same data to give a further clear understanding of the performance of our Sampling Markov Method. But in using these three methods, the traffic flow is assumed to be complete, that is, without missing data. From these experiments we can see the performance divergence of forecasting with incomplete data and forecasting with complete data. The forecasting errors evaluated by root mean square error (RMSE) are listed in Table 1 where the first line denotes the road sections, e.g. Lmq is the abbreviation of Liangmaqiao.

For vector $X = [x_1, x_2, \dots, x_n]$ and its estimation $Y = [y_1, y_2, \dots, y_n]$, the performance measure RMSE can be given in the following form:

$$RMSE(X, Y) = \left[\frac{\sum_{i=1}^n (y_i - x_i)^2}{n} \right]^{1/2}.$$

From the results, we can see that for each of the four road sections, the forecasting RMSE of the Sampling Markov Chain method is always the second least. In other words, the performance of our Sampling Markov Chain method is superior to the Joint Distribution Recalculating method and the Historical Average Markov Chain method for short-term traffic flow with incomplete data. Furthermore, its performance is even superior to the Historical Average and the Random Walk methods for traffic flow forecasting with intact data. Through careful observation, we can still find that the performance divergence of our Sampling Markov Chain method (forecasting on the INCOMPLETE data) with the Markov Chain Model method (forecasting on the COMPLETE data) is quite small.

Although our method is based on the statistical techniques, it is still very efficient because the parameter learning of GMM can be done in advance. Besides, with the ever-increasing capability of digital computers coupled with

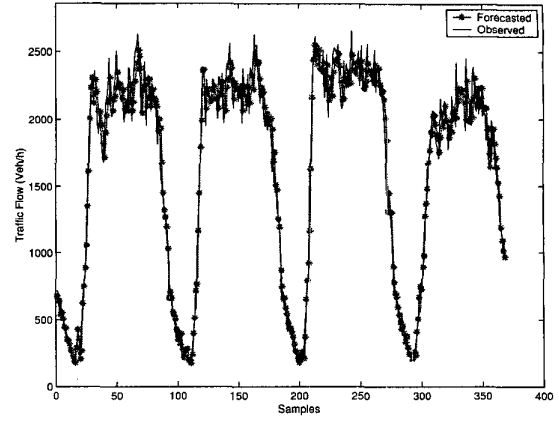


Fig. 3. Forecasted flow vs. observed flow

TABLE I
FORECASTING ERRORS OF DIFFERENT METHODS

	Lmq	Tns	Zxg	Slt
Historical Average	216.41	210.32	318.88	389.53
Random Walk	169.79	117.97	258.13	257.53
Markov Chain Model	144.19	101.76	206.06	224.64
Joint Distribution Recalculating	154.61	115.08	221.98	256.20
Historical Average Markov Chain	158.20	147.36	232.40	271.83
Sampling Markov Chain	152.97	107.98	218.82	246.88

a simultaneous decrease in the cost, the computation cost will not be a heavy burden. At the same time, the statistical characteristic of our method also brings along an appealing feature, that is, given more historical data to training, the estimated parameters of GMM will be more accurate, and the forecasting performance would be even better.

IV. CONCLUSION

In this paper, Sampling Markov Chain method is proposed to carry out short-term traffic flow forecasting with incomplete data. Gaussian Mixture Model and Competitive EM algorithm are applied to estimate the state transition probability density function of Markov Chain. The idea of Monte Carlo integration is used to approximate the missing or unavailable (incomplete) data. The experimental results with vehicular traffic flow show that our approach can achieve much better performance than several other methods.

However, there is still much work to do in the future, three of which are listed below.

- 1). how to evaluate the relationship between the number of the sampling points and the final forecasting performance.
- 2). how to expand the proposed method to forecast the traffic flow in road network in case of incomplete data.
- 3). How to apply the Sampling Markov Chain method to other kind of time series forecasting problems (e.g. travel speed of high way, traffic of broadband ISDN), etc.

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